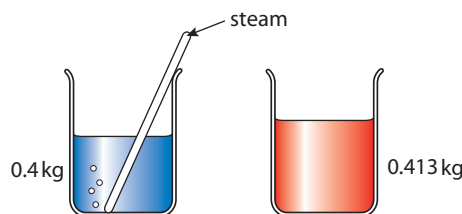


Latent heat of vaporization of water

To measure the latent heat of vaporization, steam is passed into cold water. Some of the steam condenses in the water, causing the water temperature to rise. The heat from the steam = the heat to the water.

Figure 3.17 By measuring the rise in temperature, the specific latent heat can be calculated.



In Figure 3.17, 13 g of steam have condensed in the water, raising its temperature by 20 °C. The steam condenses then cools down from 100 °C to 30 °C.

$$\text{Heat from steam} = ml_{\text{steam}} + mc\Delta T_{\text{water}}$$

$$0.013 \times L + 0.013 \times 4.18 \times 10^3 \times 70 = 0.013L + 3803.8$$

$$\begin{aligned} \text{Heat transferred to cold water} &= mc\Delta T_{\text{water}} = 0.4 \times 4.18 \times 10^3 \times 20 \\ &= 33\,440 \text{ J} \end{aligned}$$

Since heat from steam = heat to water

$$0.013L + 3803.8 = 33\,440$$

$$\text{So } L = \frac{33\,440 - 3803.8}{0.013}$$

$$L = 2.28 \times 10^6 \text{ J kg}^{-1}$$

Heat loss

In both of these experiments, some of the heat coming from the hot source can be lost to the surroundings. To reduce heat loss, the temperatures can be adjusted, so you could start the experiment below room temperature and end the same amount above (e.g. if room temperature is 20 °C, then you can start at 10 °C and end at 30 °C).

Transfer of water

In the specific heat capacity experiment, droplets of hot water may be transferred with the metal block. This would add extra energy to the water, causing the temperature to rise a little bit too high. In the latent heat experiment, droplets of water sometimes condense in the tube – since they have already condensed, they don't give so much heat to the water.

3.3 Kinetic model of an ideal gas

Assessment statements

- 3.2.9 Define *pressure*.
- 3.2.10 State the assumptions of the kinetic model of an ideal gas.
- 3.2.11 State that temperature is a measure of the average random kinetic energy of the molecules of an ideal gas.
- 3.2.12 Explain the macroscopic behaviour of an ideal gas in terms of a molecular model.

The ideal gas

Of the three states of matter, the gaseous state has the simplest model; this is because the forces between the molecules of a gas are very small, so they are able to move freely. We can therefore use what we know about the motion of particles learnt in the mechanics section to study gases in more detail.

According to our simple model, a gas is made up of a large number of perfectly elastic, tiny spheres moving in random motion.

This model makes some assumptions:

- The molecules are perfectly elastic.
- The molecules are spheres.
- The molecules are identical.
- There are no forces between the molecules (except when they collide) – this means that the molecules move with constant velocity between collisions.
- The molecules are very small, that is, their total volume is much smaller than the volume of the gas.

Some of these assumptions are not true for all gases, especially when the gas is compressed (when the molecules are so close together that they experience a force between them). The gas then behaves as a liquid. However, to keep things simple, we will only consider gases that behave like our model. We call these gases ideal gases.



Temperature of a gas

From our general particle model of matter, we know that the temperature of a gas is directly related to the average KE of the molecules. If the temperature increases, then the speed of the particles will increase.

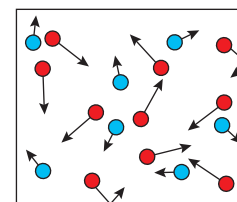
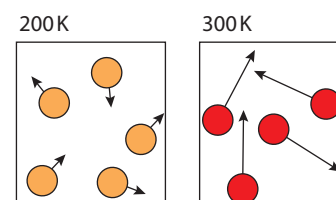


Figure 3.18 Simple model of a gas in a box. In reality the molecules have a range of velocities, not just two.

Nitrogen becomes a liquid at low temperatures.

Figure 3.19 The molecules in a hot gas have a higher average KE.



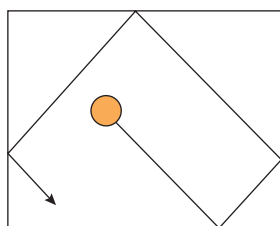


Figure 3.20 A rubber ball bouncing around a box.

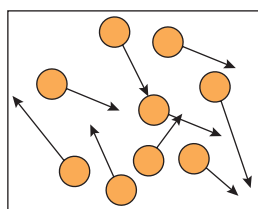


Figure 3.21 Many rubber balls bouncing around a box.

The atmosphere also exerts a pressure; this changes from day to day but is approximately 100 kPa.
1 pascal = 1 Pa = 1 N m⁻²

Figure 3.22 A gas in a piston can be used to vary the properties of a gas.

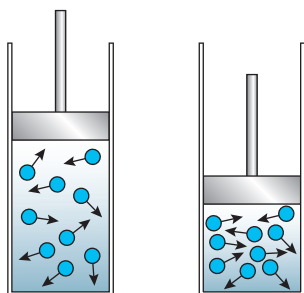


Figure 3.23 The volume of a gas is reduced.

Pressure of a gas

Let us apply what we know about particles to one molecule of a gas. Consider a single gas molecule in a box. According to the model, this is like a perfectly elastic sphere bouncing off the sides

We can see that this particle keeps hitting the walls of the container. Each time it does this, its direction, and therefore its velocity, changes.

Newton's first law of motion says that if a particle isn't at rest or moving with a constant velocity then it must be experiencing an unbalanced force. The particle is therefore experiencing an unbalanced force.

Newton's second law says that the size of this force is equal to the rate of change of momentum, so the force will be greater if the particle travels with a greater speed, or hits the sides more often.

Newton's third law says that if body A exerts a force on body B, then body B will exert an equal and opposite force on A. The wall exerts a force on the particle, so the particle must exert a force on the wall.

If we now add more molecules (as in Figure 3.21) then the particles exert a continuous force, F , on the walls of the container. If the walls have a total area A , then since

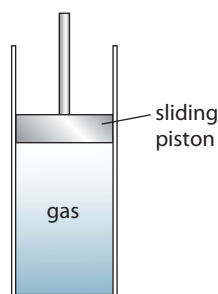
$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

we can say that the pressure exerted on the walls is F/A — in other words, the particles exert a pressure on the container.

It is important to realise that we have been talking about the gas model, not the actual gas. The model predicts that the gas should exert a pressure on the walls of its container and it does.

Properties of a gas

We can now use the particle model to explain why a gas behaves as it does.

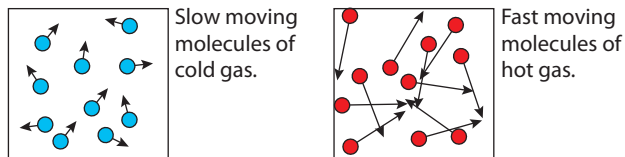


If you push on the piston you can feel the gas push back.

Pressure and volume

If the volume is reduced, the particles hit the walls more often, since the walls are closer together. The force exerted by the molecules is equal to the rate of change of momentum; this will increase if the hits are more frequent, resulting in an increased pressure.

Pressure and temperature



Increase in temperature increases the speed of the molecules. When the molecules hit the walls, their change of momentum will be greater and they will hit the walls more often. The result is a greater rate of change of momentum and hence a larger force. This results in an increase in pressure.

Doing work on a gas

When you push the piston of a pump, it collides with the molecules, giving them energy (rather like a tennis racket hitting a ball). You are doing work on the gas. The increase in kinetic energy results in an increase in temperature and pressure. This is why the temperature of a bicycle pump increases when you pump up the tyres.

Gas does work

When a gas expands, it has to push away the surrounding air. In pushing the air away, the gas does work, and doing this work requires energy. This energy comes from the kinetic energy of the molecules, resulting in a reduction in temperature. This is why an aerosol feels cold when you spray it; the gas expands as it comes out of the canister.

Figure 3.24 The temperature of a gas is increased.



W To understand how pressure, temperature and volume of a gas are related, visit heinemann.co.uk/hotlinks, enter the express code 4426P and click on Weblink 3.2.

3.4 Thermodynamics

Assessment statements

- 10.1.3 Describe the concept of the absolute zero of temperature and the Kelvin scale of temperature
- 10.1.1 State the equation of state for an ideal gas.
- 10.1.4 Solve problems using the equation of state of an ideal gas.
- 10.1.2 Describe the difference between an ideal gas and a real gas.

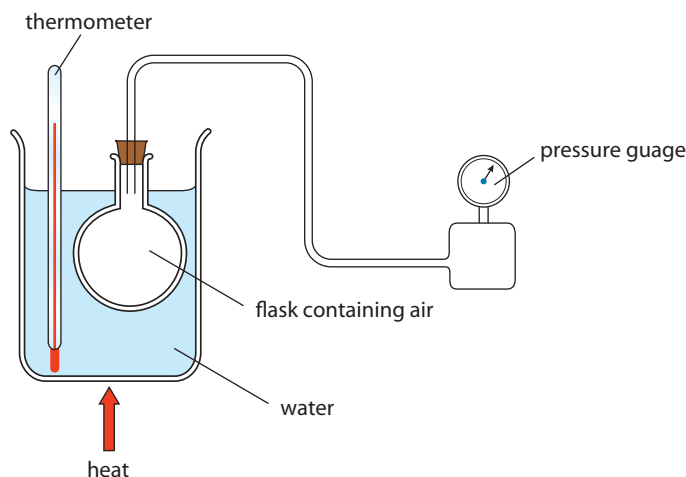


Figure 3.25 The pressure of a fixed volume of gas can be measured as the temperature is changed.